Inode

File System

Group of files and relevant information stored logically

Eg-unix file system, windows file system

In unix everything is treated as a file -it may be application, utility, directory, devices

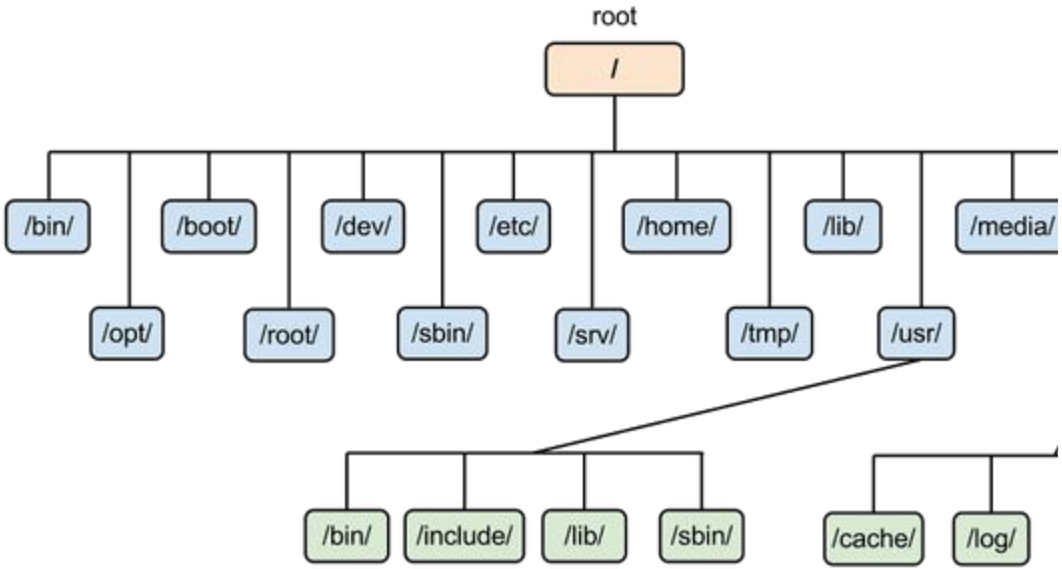
Unix file system is similar to tree data structure

Directories keep the relate files together and separate from other group of related files

Top of the tree is directory called as root denoted by slash (/)

All files/directories are contained in root directory

Home directory tilde (tilde (~)



bin -commands

home-user names-

dev-device

/tmp-temp

Path

Absolute Path-name

An absolute path is defined as the specifying the location of a file or directory from the root directory(/).
Start at the root directory (/) and work down.

- Write a slash (/) after every directory name (last one is optional)
- `$cat abc.sql`
- will work **only** if the fie **“abc.sql”** exists in your current directory.
- However, if this file is not present in your working directory and is present somewhere else say in /home/kt , then this command will work only if you will use it like shown below:

  `cat /home/kt/abc.sql`
- *An **absolute path** is defined as specifying the location of a file or directory from the root directory(/). In other words,we can say that an absolute path is a complete path from start of actual file system from / directory.*
- **Relative path**

  Relative path is defined as the path related to the present working directly(pwd). It starts at your current directory

Inode

Index node

Inode table- Is a data structure that contains information about files stored in a inode table

Each file is represented by an inode. Each file has entry in inode table’and this entry contains meta data

To identify different file we assign unique number to each file and that number is called inode number

To find the inode number of each file

```
[root@stackscale ~]$ ls -i /var/log/lastlog

   17381397 /var/log/lastlog
```

Inode

An inode entry takes 64 bytes

<table>
<tr><td rowspan="10">64 bytes</td><td>mode</td><td># of hard links</td></tr>
<tr><td>uid</td><td>gid</td></tr>
<tr><td colspan="2">size (in bytes)</td></tr>
<tr><td colspan="2">accessed time</td></tr>
<tr><td colspan="2">modification time</td></tr>
<tr><td colspan="2">creation time</td></tr>
<tr><td colspan="2">7 direct zone pointers</td></tr>
<tr><td colspan="2">1 indirect zone pointer</td></tr>
<tr><td colspan="2">1 doubly indirect zone pointer</td></tr>
<tr><td colspan="2">1 triply indirect zone pointer</td></tr>
</table>

Storage of file

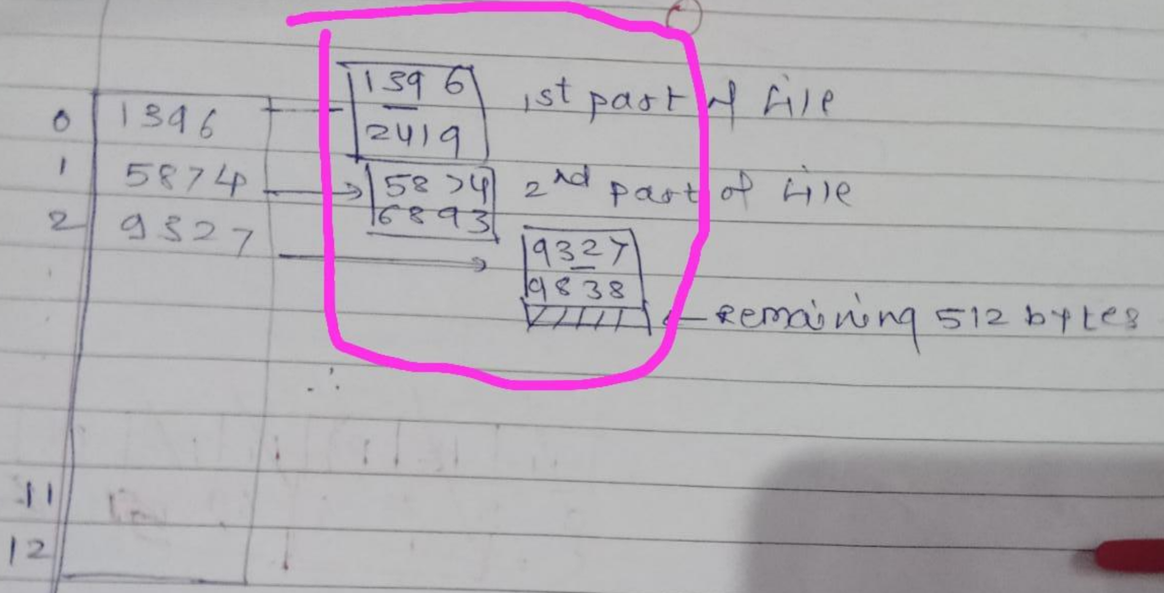
Inode entry contains addresses of the blocks where file is physically present

Eg -consider block of 1 KB-1024 bytes

There are 13 entries in the inode entry of a file-means we can store 13 addresses in inode.

Entries are numbered from 0 to 12

First 10 entries (0-9) point to the 1KB block where the actual file contents are stored. So these entries directly provide us the addresses at which the file is being stored size of the file 2560 bytes



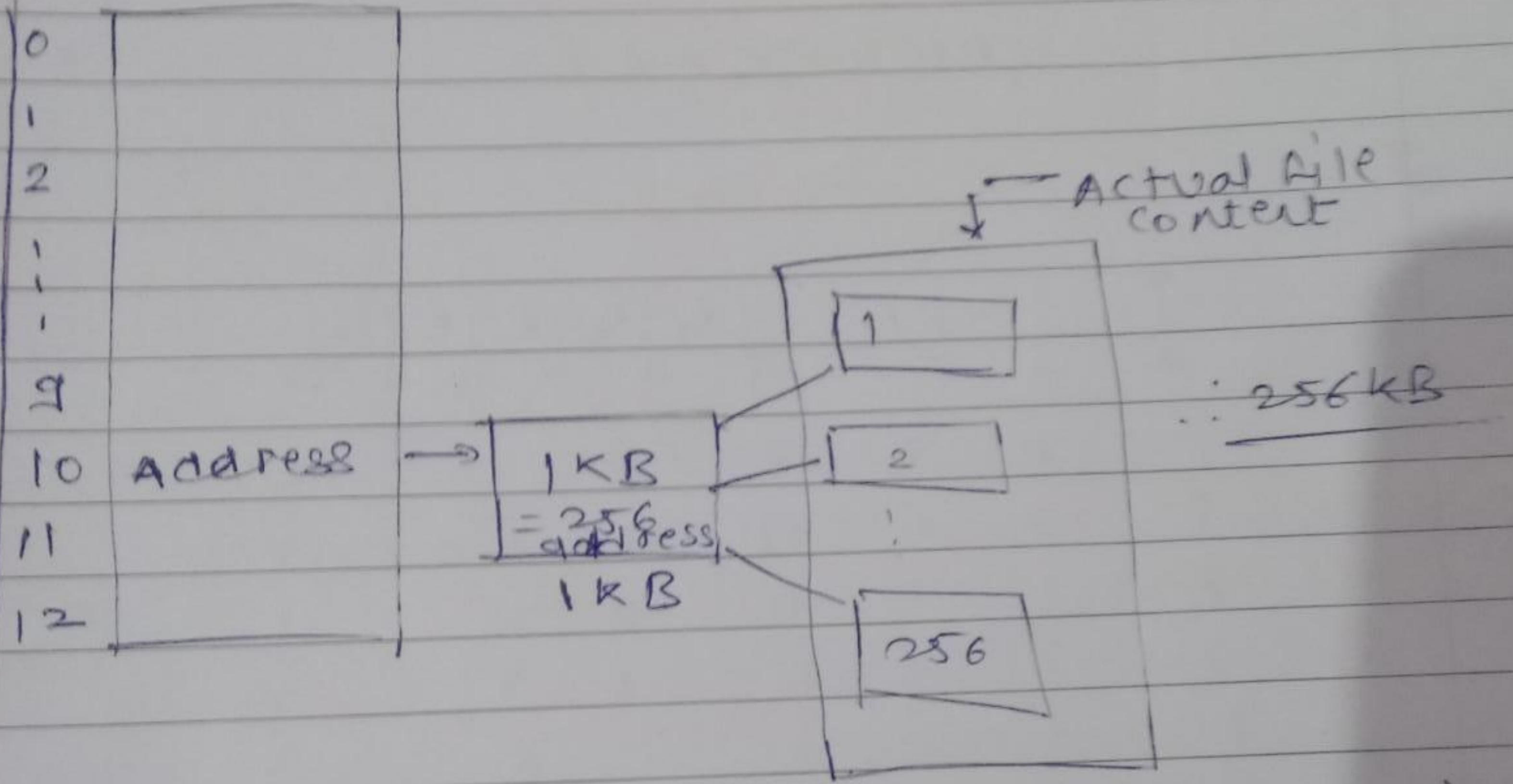
1396 start addr
+ 1023
2419 - last addr

5874	1024	2560
+ 1023	+ 1024	- 2048
6897	2048	0512

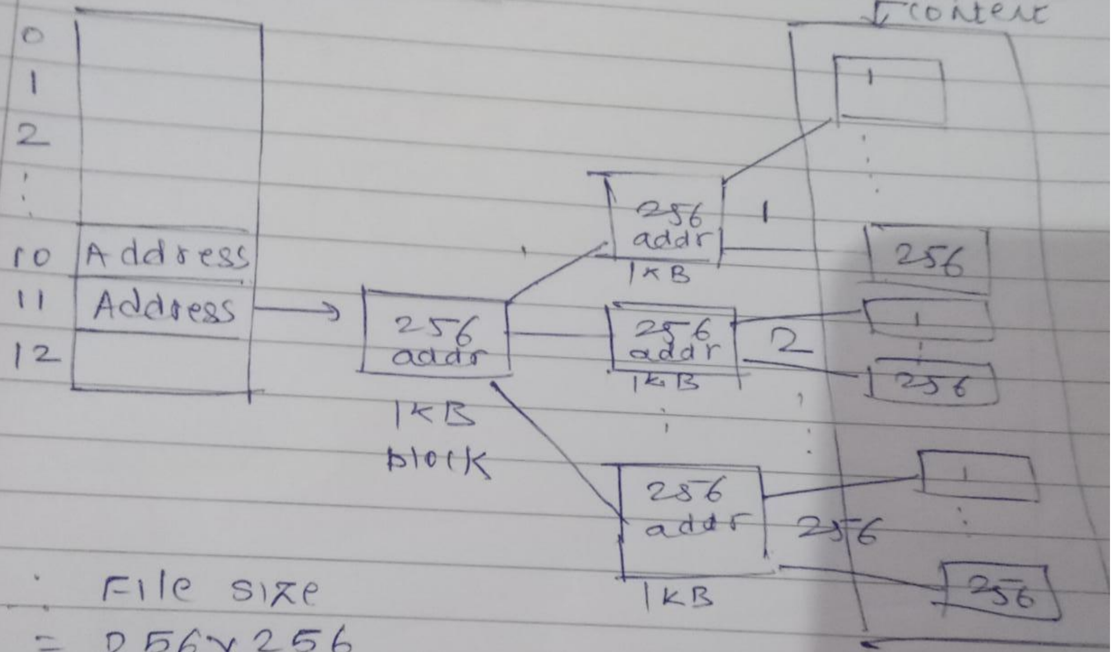
9327

File size of max 10KB can be stored from 0-9

Single Indirection



Double Indirection



$\therefore$ File size

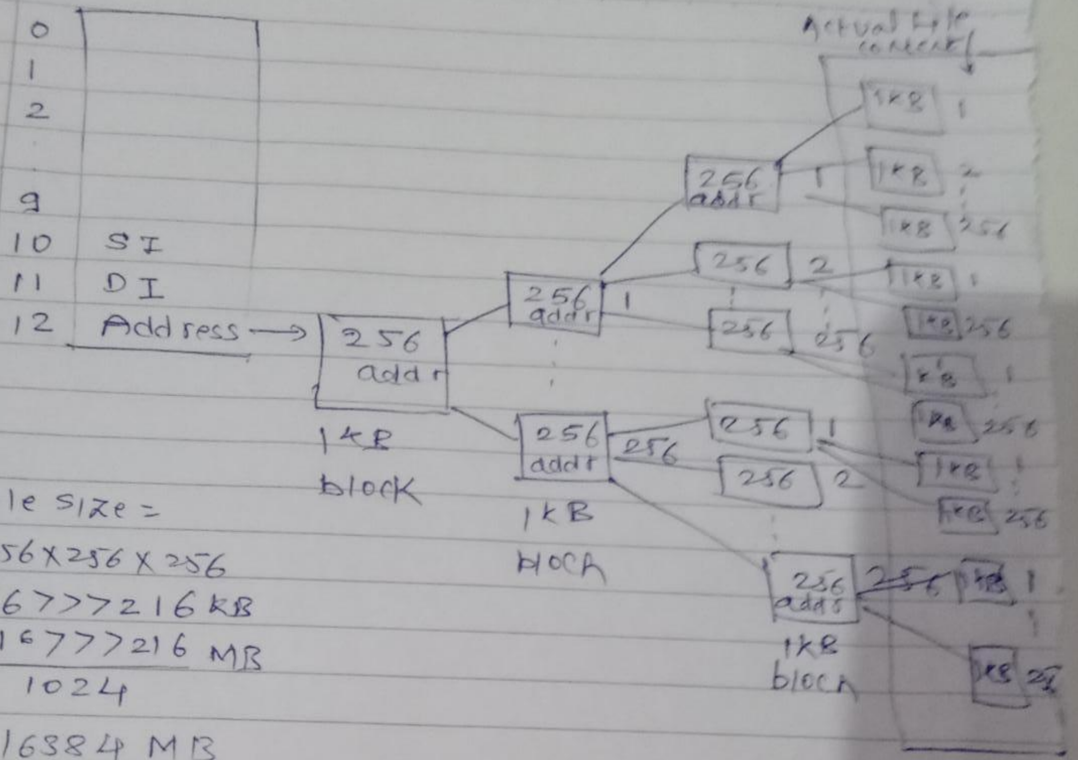
$= 256 \times 256$

$= 65536 \text{ kB}$

$= \frac{65536}{1024} \text{ MB}$

$= 64 \text{ MB}.$

Triple Indirect



File size =

$256 \times 256 \times 256$

$= 16777216 \text{ KB}$

$= \dfrac{16777216}{1024} \text{ MB}$

$= 16384 \text{ MB}$

$= \dfrac{16384}{1024} = 16 \text{ GB}$

Maximum file size

10kB + 256kB + 64MB + 16GB

0 — 9 10th entry 11th entry 12th entry

SI DI TI

A file system uses unix inode data structure which contains 8 direct block address, one indirect, one double indirect and one triple indirect block. The block size is 128 byte and size of each block address is 8 byte. Find the maximum possible file size.

(8+16+(16*16)+(16*16*16))*128

=8(1+2+(2*16)+(2*16*16))*128

=(3+32+512)*8*128

=547KB

A file system with 300GB disc uses a file descriptor with 8 direct block addresses, 1 indirect,1 double indirect block. Each disc block size is 128 bytes and size of each disk block address is 8 bytes. The maximum possible file size in this file system is?

Disc block size= 128 bytes

Disc block address = 8 bytes

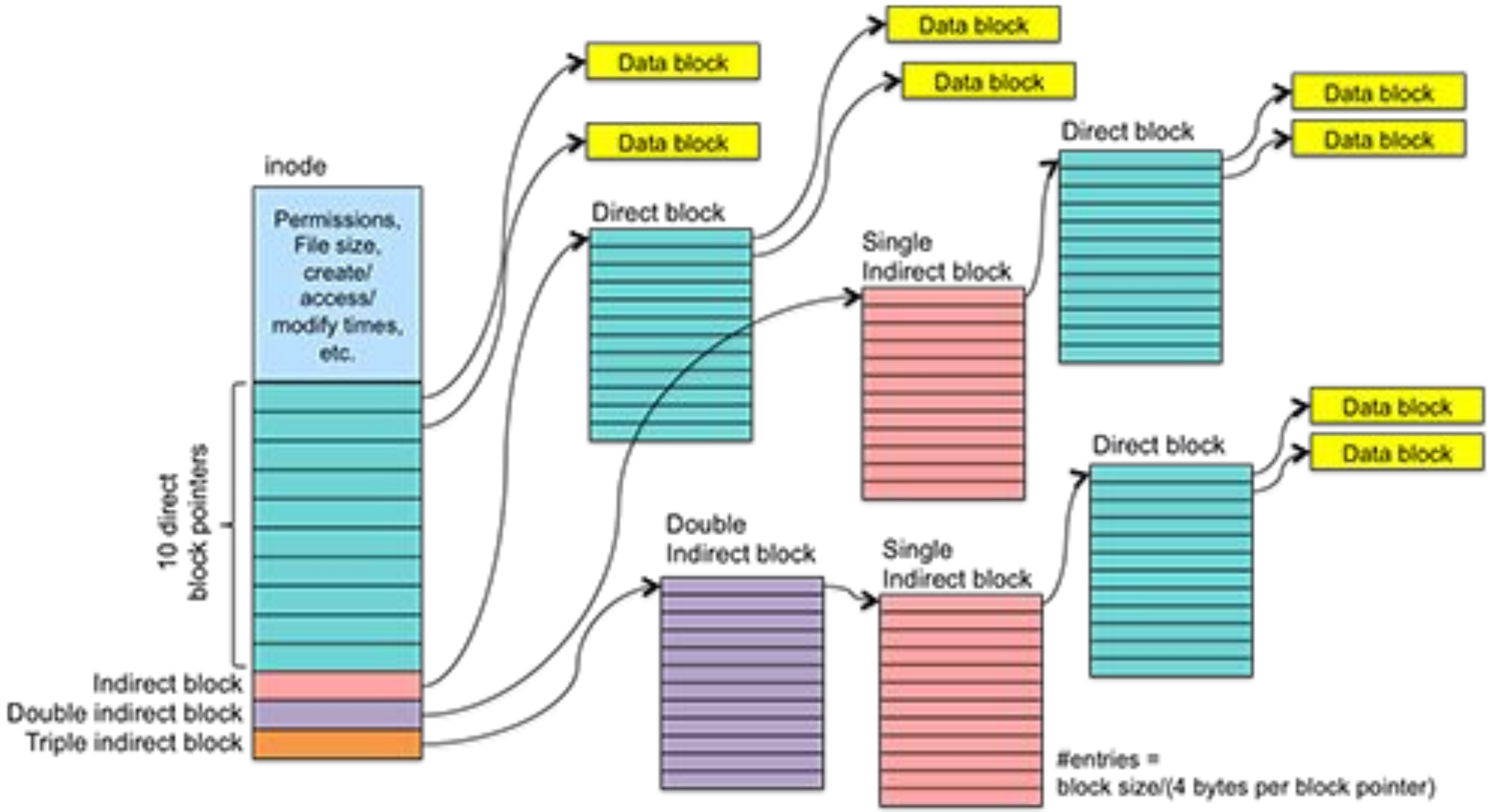
No of address=128 B/8B=16

Maximum File size=(8+16+(16*16))*128

=8(1+2+(2*16)*128

=35KB

Suppose file size 2560 bytes



Inode Block pointer